

# Does Economic Growth Translate into Human Development?

An Empirical Statistical Analysis of Economic Prosperity, Inequality, Poverty, and Human Development Across Countries

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## Abstract

This study examines whether economic prosperity is associated with human development across countries, and how inequality and poverty relate to this association. Using the 2022 cross-section of the UNDP Human Development Index (HDI, N=193 countries) merged with World Bank data on income, the Gini index, and poverty headcount ratios, we estimate ordinary least squares regressions of HDI on log income per capita, then on income together with inequality and poverty. Log income per capita is a strong, precisely estimated predictor of HDI (beta = 0.101, 95% CI [0.096, 0.106],  $R^2 = 0.90$ , N = 186), a result that is stable across an alternative year (2015), alternative samples, and alternative functional forms. Conditional on income, the Gini index is negatively associated with HDI but only borderline significant ( $p \sim 0.05$ ) and loses significance once poverty is included ( $p \sim 0.53$ ), while poverty remains a strong, significant negative predictor ( $p < 0.001$ ). A residual-based analysis identifies countries with HDI substantially above or below their income-based prediction, a ranking that is model-conditional and shown to be robust to an alternative functional form. Income shows a broadly similar-strength association with life expectancy and expected years of schooling analysed separately ( $R^2 \sim 0.64$ - $0.66$  each). All models use observational, cross-sectional data; results are interpreted as statistical associations, not causal effects, throughout. Coverage limitations in the inequality and poverty data, which are not missing at random, are a central limitation of this study.

## 1. Introduction

Gross Domestic Product (GDP) and Gross National Income (GNI) per capita remain the default proxies for national economic progress, yet they measure the volume of market production rather than how its proceeds are distributed or whether they translate into longer, healthier, better-educated lives. The UNDP's Human Development Index (HDI) was created explicitly to address this gap, combining a health dimension (life expectancy), an education dimension (expected and mean years of schooling), and a living standards dimension (income) into a single composite measure. This study asks: to what extent is economic prosperity associated with human development, and how does inequality relate to this relationship? It investigates this question using official, internationally comparable statistics, distinguishing explicitly between association, prediction, and causal claims.

## 2. Background

A large development-economics literature documents a strong cross-country correlation between income and composite development indices, alongside persistent debate over whether income growth is sufficient for human development or whether distributional factors -- inequality and poverty -- carry independent

explanatory power. Because HDI is a composite index constructed in part from income itself, and in part from health and education indicators that are also plausibly influenced by income, disentangling these relationships requires care: this study treats health and education as separate outcomes analysed on their own terms (never as predictors of HDI, of which they are components), and treats all reported relationships as associations rather than causal effects given the observational, cross-sectional design.

### 3. Data

Data come from official sources: the UNDP Human Development Report 2023/24 Statistical Annex (HDI, 193 countries, 2022), World Bank indicators for GDP, population, and the Gini index, and the World Bank Poverty and Inequality Platform for poverty headcount ratios. Because the analysis environment used for this project could not reach the UNDP/UN/World Bank sites directly, all series were retrieved through actively-maintained, citation-preserving mirrors that redistribute the same official indicators unmodified (full URLs, indicator codes, and retrieval dates in *data/raw/SOURCES.md*). Country names were harmonized to ISO3 codes via a documented crosswalk; no countries were dropped without being logged. Gini and poverty are household-survey based and reported irregularly, so each was matched to the nearest available survey year within a five-year window of the target year, with the exact source year retained for transparency. Coverage in the 2022 cross-section is high for income (96%) but incomplete for Gini (134/193, 69%) and poverty (104/193, 54%) -- missingness that is not random with respect to country characteristics (poorer, more data-scarce states are disproportionately affected), a limitation carried through every model using these variables. The complete variable-level data dictionary is in *data/processed/data\_dictionary.md*.

### 4. Methodology

The primary unit of analysis is the country in the 2022 HDI cross-section. HDI is regressed on log income per capita (log transform justified by the heavy right-skew of raw income and by the standard diminishing-marginal-returns rationale, confirmed empirically in Section 5), then on income together with the Gini index, and finally with poverty added. Heteroskedasticity-robust (HC3) standard errors are used for formal inference given evidence of mild heteroskedasticity. Multicollinearity is assessed via Variance Inflation Factors; influential observations via Cook's distance and leverage. The flagship analysis computes residuals from the income-only model and interprets them as "higher/lower than income-based prediction" -- explicitly not as causal efficiency or policy performance. Robustness is assessed via an alternative year (2015), an alternative sample (excluding micro-states and influential observations), and an alternative functional form (adding a quadratic income term).

### 5. Results

**Income and HDI.** Log income per capita alone explains a large share of cross-country variation in HDI ( $R^2 = 0.90$ ,  $N = 186$ ), with a highly precise, positive coefficient (beta = 0.101,  $p < 10^{-90}$ ). Figure 1 shows this relationship.

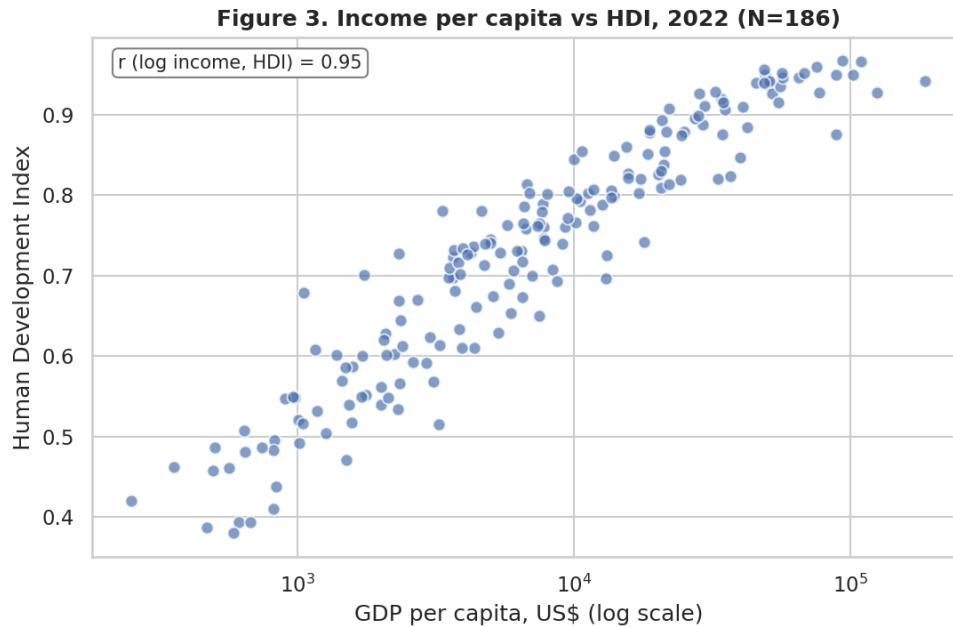


Figure 1. Income per capita (log scale) vs HDI, 2022.

**Adding inequality and poverty.** Conditional on income, the Gini index is negatively associated with HDI (beta = -0.0013,  $p \sim 0.05$ ,  $N = 133$ ) -- borderline significant. Once poverty is added ( $N = 103$ ), the Gini coefficient's association with HDI is no longer statistically distinguishable from zero ( $p \sim 0.53$ ), while poverty remains a strong, highly significant negative predictor (beta = -0.0017,  $p < 0.001$ ). Table 1 summarizes.

| Model            | Income beta | Gini beta | Poverty beta | N   | R2    |
|------------------|-------------|-----------|--------------|-----|-------|
| M1: Income only  | 0.101       | —         | —            | 186 | 0.896 |
| M2: + Inequality | 0.103       | -0.0013   | —            | 133 | 0.911 |
| M3: + Poverty    | 0.094       | -0.0004   | -0.0017      | 103 | 0.928 |

Table 1. Model comparison, 2022 cross-section.

**Heterogeneity and relative performance.** Substantial HDI variation remains among countries at similar income levels (Figure 2). A residual-based ranking (observed minus income-predicted HDI, Figure 3) identifies Tajikistan, Sri Lanka, Kyrgyzstan, and Uzbekistan as having the largest positive residuals, and Djibouti, Liechtenstein, Qatar, and several fragile states as having the largest negative residuals. This ranking correlates highly (Spearman) with the ranking from an alternative quadratic-income specification, indicating it is not an artifact of the specific functional form chosen -- but it remains conditional on this model's specification and should not be read as an absolute judgment of national performance.

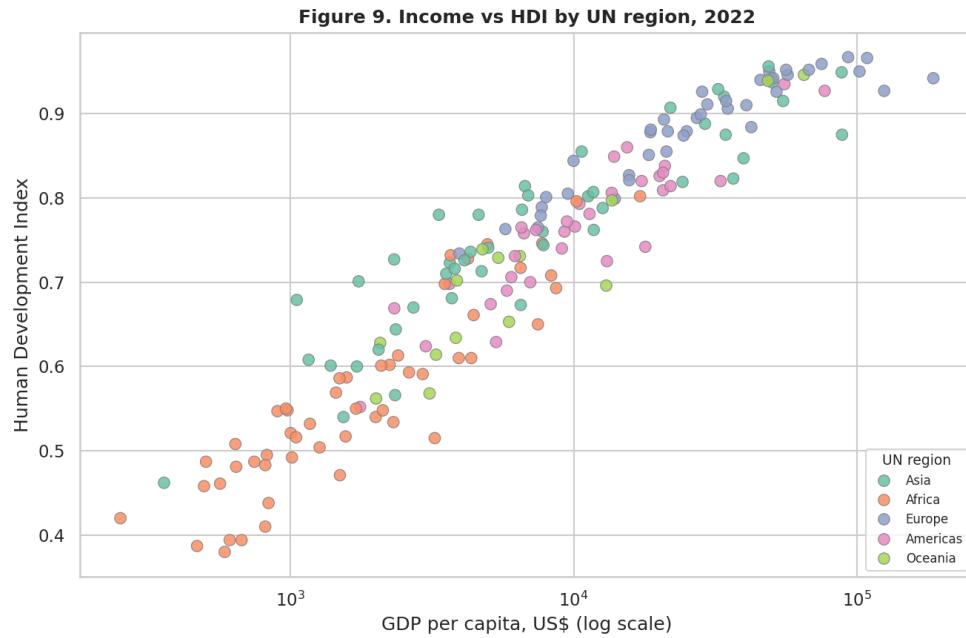


Figure 2. Income vs HDI by UN region, 2022.

Figure 14. Relative human-development performance ranking (Model 1 residuals)

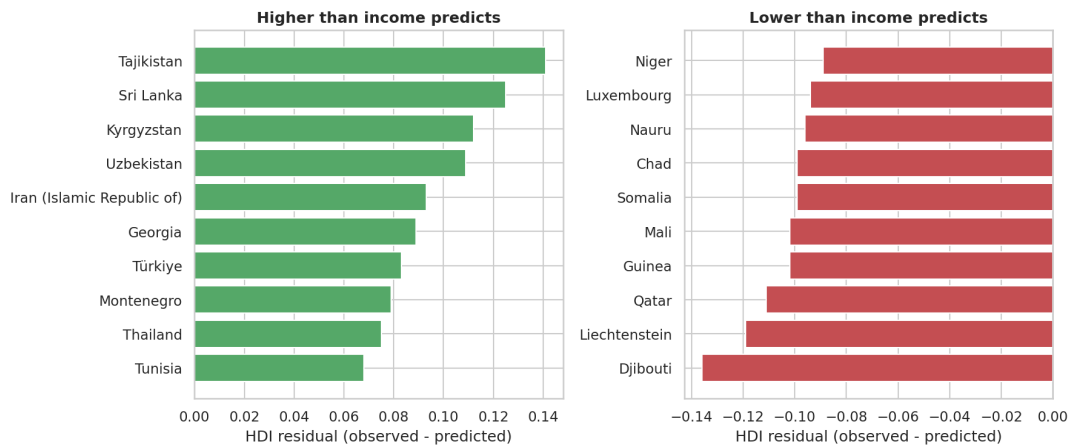


Figure 3. Countries with the largest positive and negative HDI residuals relative to Model 1's income-based prediction.

**Development dimensions.** Analysed separately (never as predictors of HDI, of which they are components), income is significantly associated with both life expectancy ( $R^2 = 0.66$ ,  $N = 179$ ,  $p < 10^{-40}$ ) and expected years of schooling ( $R^2 = 0.64$ ,  $N = 182$ ,  $p < 10^{-38}$ ), of broadly similar strength -- though this comparison is limited by a data-vintage mismatch (these series reflect data through 2013/2017, matched against 2022 income figures).

## 6. Robustness and Sensitivity

The income-HDI association is stable across every robustness check performed: a different year (2015:  $\beta = 0.102$ ,  $R^2 = 0.873$ ,  $N = 190$ , essentially identical to 2022), a sample excluding micro-states, a sample excluding the most influential observations (Cook's distance above the  $4/n$  threshold),

heteroskedasticity-robust standard errors, and an alternative functional form (adding a quadratic income term, which improves fit only marginally:  $R^2 = 0.905$  vs.  $0.896$ ). By contrast, the inequality (Gini) association is the least robust finding in this project: directionally consistent but small in magnitude, only borderline significant in isolation, and not distinguishable from zero once poverty is included in the model. This asymmetry in robustness is itself a substantive finding, and is reported as such rather than smoothed over.

## 7. Discussion

These results are consistent with the well-established cross-country income-development gradient, and extend it by showing that (a) this gradient is highly stable across reasonable modelling choices, (b) a poverty measure appears to carry more independent information about HDI shortfalls than the Gini coefficient once both are considered jointly, and (c) meaningful, specification-robust heterogeneity exists around the income-HDI relationship that a single-predictor model does not capture. None of this establishes that income growth causes human development gains: omitted variables (institutions, governance, historical legacies), reverse causality (healthier and more educated populations are also more economically productive), and non-random missingness in the inequality and poverty data are all plausible alternative explanations for the patterns observed, and cannot be ruled out with this observational, cross-sectional design.

## 8. Limitations

- Observational, cross-sectional design: no causal claims can be supported.
- Gini and poverty coverage is incomplete and not missing at random (skews toward poorer, more data-scarce countries).
- Life expectancy and schooling reflect an older data vintage (2013/2017 or earlier) than the 2022 income and HDI figures used elsewhere.
- GDP per capita is used in place of GNI per capita (HDI's own income measure) because a suitable GNI series could not be retrieved through an accessible source.
- No World Bank income-group classification was available through an accessible mirror; UN region/sub-region is used instead.
- HDI and the Gini index are both single summary statistics with well-known conceptual limitations (equal dimension weighting; insensitivity to the full distribution).
- The residual-based country ranking is model-dependent and should not be read as an absolute judgment of national development performance.

## 9. Conclusion

Economic prosperity is strongly and robustly associated with human development across countries: log income per capita alone explains roughly 90% of cross-country variance in HDI in 2022, a relationship stable across years, samples, and functional forms. Inequality shows a directionally consistent but weak and fragile association with HDI once poverty is accounted for, while poverty itself remains a robust, independent predictor. Considerable, specification-robust heterogeneity exists among similarly-wealthy countries, motivating further research into the institutional, historical, and policy factors this income-only model cannot capture. All findings are presented as statistical associations, consistent with the observational nature of the underlying data.

## References

- UNDP Human Development Report Office (2024). *Human Development Report 2023/24*, Statistical Annex. <https://hdr.undp.org/data-center>
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